The best defense is a good offense

A pragmatic path to **Continuous Purple Teaming**



Cas van Cooten

ONE Conference 2026

Who am I?

- Long-time offensive security enthusiast & Red Teamer
- Conference speaker on offensive security and building security tools
- Former: External Red Team @ Deloitte → internal Red Team @ ABN AMRO
- Current: Co-founder of **Offensys**



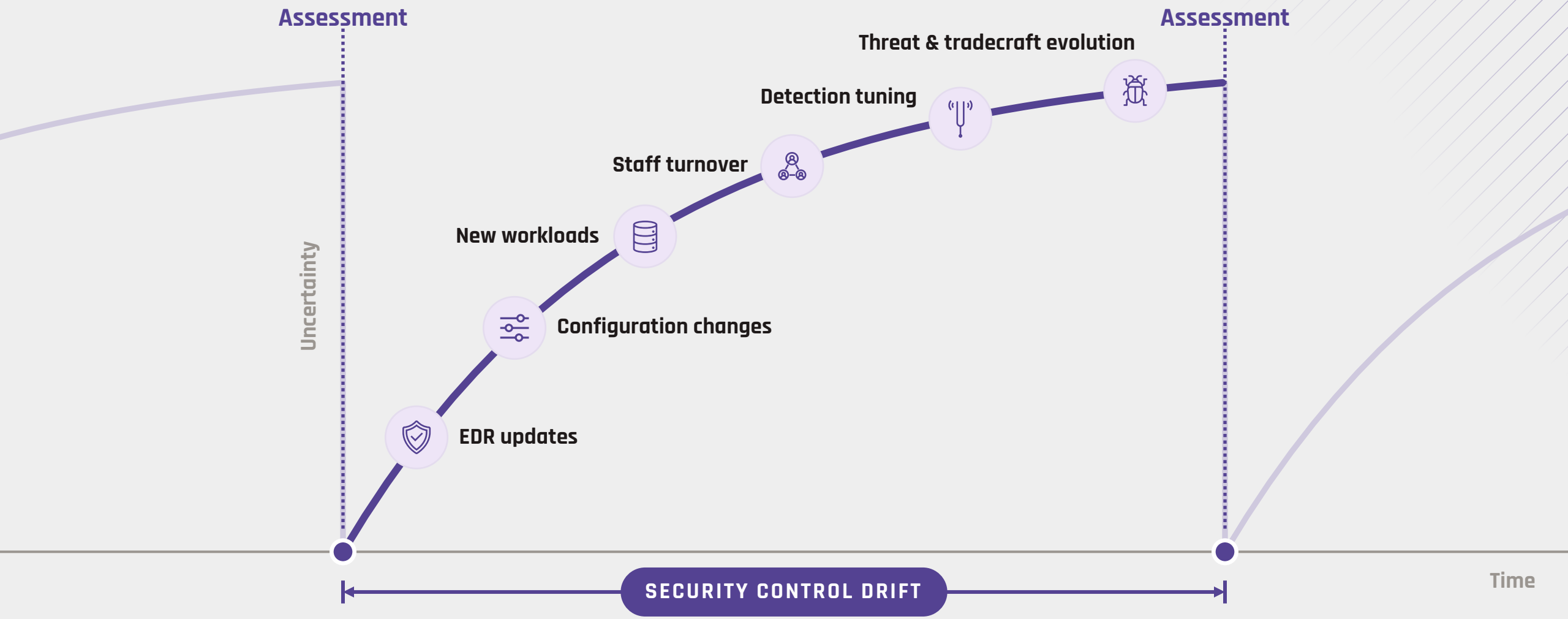
Cas van Cooten

Offensive Security Specialist,
Co-Founder & Tech Lead

Attackers iterate in hours.
You validate once a year.

The problem with point-in-time assessments

Between two assessments, every change can silently break a control that you believed worked



Not just an operational pain



SECURITY EXECUTIVES

"How do I prioritize what to fix, and how do I measure and prove the ROI of what we've invested in?"



SECURITY OPERATIONS / MSSP

"How do I know my controls didn't regress since last week?"
"How do I hold my MSSP to a measurable standard?"
"How do I efficiently engineer a detection for a new technique?"

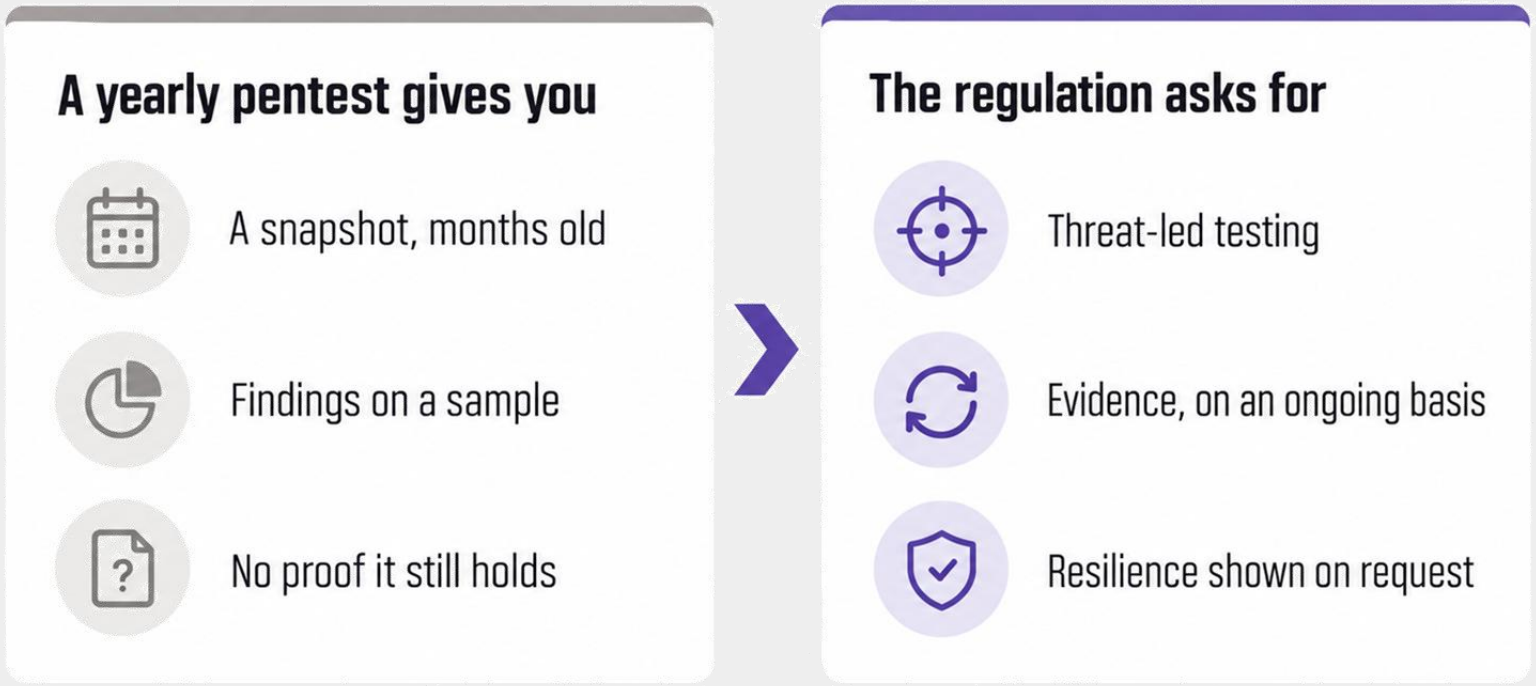


OFFENSIVE SECURITY

"How do I scale my expertise and stop manually repeating the same simulations across every engagement?"

Regulators now ask for proof, not a report

DORA and NIS2 expect you to demonstrate that controls work now, not that they were tested once.



See: DORA Art. 24-26, NIS2 Art. 21

What is Purple Teaming?



Red Team

The Offense

Emulates adversary TTPs to expose exploitable gaps across people, process, and technology.

THIS TALK



Purple Team

The Collaboration

Offense and Defense collaborating in full transparency: red runs simulations and provides offensive insight, while blue hunts and improves control posture.



Blue Team

The Defense

Prevents, detects, and responds, continuously improving identify gaps and turn them into measurable gains in visibility and control.

So how do we improve?

The Gartner Solution:

Adversarial Exposure Validation (AEV)

Technologies that deliver **consistent, continuous and automated** evidence of the feasibility of an attack. [...] AEV provides automated execution of both simplified and/or extensible attack scenarios.

Source: Gartner, Market Guide for AEV

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The Implementation:

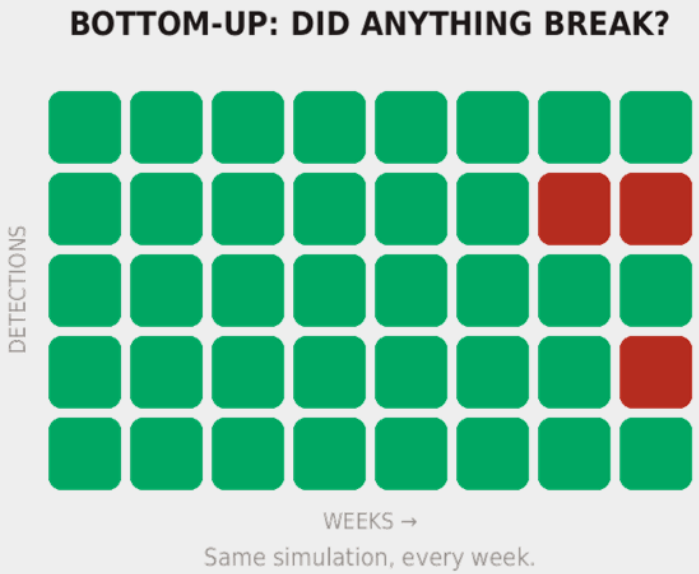
Continuous Purple Teaming (CPT)

Continuous Purple Teaming is AEV with an emphasis on insight for the blue team: every simulation feeds a **continuous feedback loop** that measurably improves prevention and detection.

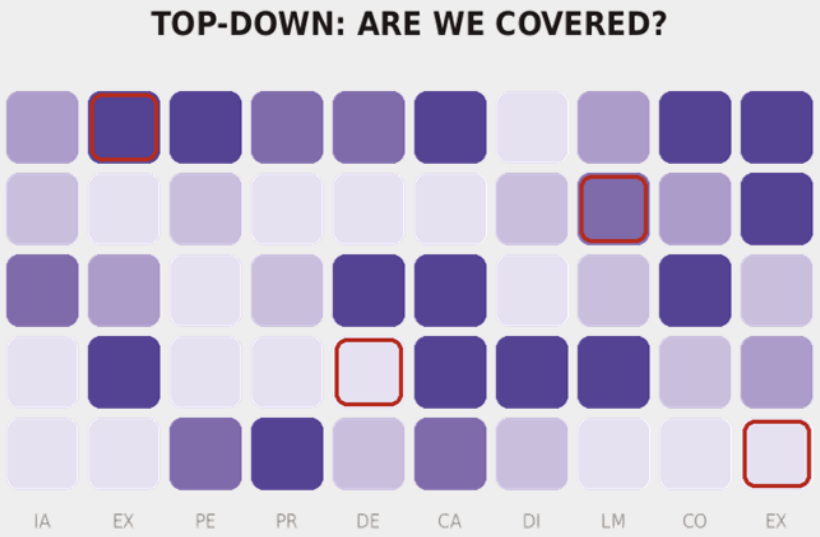
The CPT feedback loop



Simulate: Two directions of validation



Same simulation, every week.



Coverage of your threat profile on MITRE ATT&CK.

Simulate: Test behavior, not tools

Behavior-based tests take more work to build, and hold up far longer

TOOL-BASED TEST

"Do we catch mimikatz.exe?"

Effort to build	**Low** replay the known tool
Effort to evade	**Low** rename the file
A pass proves	**Narrow** one tool is caught

BEHAVIOR-BASED TEST

"Do we catch anyone stealing passwords from memory?"

Effort to build	**High** research and engineer
Effort to evade	**High** needs a new approach
A Pass proves	**Broad** the technique is caught

Simulate: Test behavior, not tools

Tool-based tests overstate what your controls stop

Alerts raised



Attacks blocked



WHAT IT MEANS

Same attacks, same EDR. Only the simulation type changed.

If your tests copy a known tool, your dashboard shows protection you don't have.

That is not bad news. The lower number is the real resilience, and the place to start improving.

Experiment: Offensys (2026), one widely deployed EDR at default settings

The CPT feedback loop



Measure: What did your controls actually do?

Every simulation gets one outcome: blocked, detected or missed

SIMULATION	LAST WEEK	THIS WEEK
Phishing payload runs	Blocked	Blocked
Password spraying on cloud accounts	Detected	Detected
Credential theft from memory	Detected	Missed
Lateral movement to a file server	Missed	Missed
Mass file encryption	Blocked	Blocked
Data theft over DNS	Missed	Missed

Blocked a control stopped it **Detected** an alert fired **Missed** nothing noticed it

Measure: What did your controls actually do?

Every simulation gets one outcome: blocked, detected or missed

SIMULATION	LAST WEEK	THIS WEEK
Phishing payload runs	Blocked	Blocked
Password spraying on cloud accounts	Detected	Detected
Credential theft from memory REGRESSION	Detected	Missed
Lateral movement to a file server	Missed	Missed
Mass file encryption	Blocked	Blocked
Data theft over DNS	Missed	Missed

Blocked a control stopped it **Detected** an alert fired **Missed** nothing noticed it

The CPT feedback loop



Prioritize: A ranked fix list

Weigh each outcome by how much that attack matters to you

SIMULATION	THIS WEEK	IMPACT FOR US
Phishing payload runs	Blocked	High
Password spraying on cloud accounts	Detected	High
Credential theft from memory	Missed	High
Lateral movement to a file server	Missed	Medium
Mass file encryption	Blocked	High
Data theft over DNS	Missed	Low

Prioritize: A ranked fix list

Weigh each outcome by how much that attack matters to you

#	SIMULATION	THIS WEEK	IMPACT FOR US	
1	Credential theft from memory	Missed	High	FIX FIRST
2	Lateral movement to a file server	Missed	Medium	FIX FIRST
3	Data theft over DNS	Missed	Low	
4	Password spraying on cloud accounts	Detected	High	
5	Phishing payload runs	Blocked	High	
6	Mass file encryption	Blocked	High	

Outcome × impact = where to spend next week

Getting started with Continuous Purple Teaming

MATURITY

1

Start manual & prioritize

Run manual simulations, picking the ones that matter most for your business. Deliver value as soon as possible.

2

Automate

Use a (custom) harness to automate your simulations and run them repeatably.

3

Measure control response

Build your own automation to measure prevention and detection automatically.

Close the loop & iterate

Ensure priority findings are followed up. Then build on CPT program to expand scope, coverage and value.

Atomic Red Team
OPEN-SOURCE LIBRARY

MITRE Caldera
OPEN-SOURCE HARNESS

Three questions to ask your team tomorrow

1. Are we really protected against the threats that matter to us?
2. Could we prove our control effectiveness to a regulator today?
3. Do the controls we implemented yesterday still work today?

If any answer is "we don't know", that is where to start

OFFENSYS

Thank you!

Find us at Expo stand 11

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ONE Conference

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